

Detect Session Hijacking

Execution mode	Authorized, isolated training lab; no public systems were targeted.
Output status	Representative/sanitized lab output recorded for submission. Replace with instructor-required screenshots if a live lab run is performed.

Aim

Detect ARP spoofing and duplicate session behavior using packet analysis and application logs.

Theory

Detection can be performed at the network layer and at the application layer. The evidence below was generated from a synthetic replay of the previous experiment and uses redacted token values.

Tools and Lab Scope

- Wireshark
- arpswatch/Xarp-style detector
- Apache access log fixture
- Snort optional

Procedure and Recorded Execution

Step 1. Capture and filter ARP - Monitor the victim-side interface and check for duplicate-address indicators.

```
display filter: arp.duplicate-address-detected  
alert: 192.168.56.1 claimed by two MAC addresses
```

Step 2. Check the ARP table - Compare repeated gateway MAC values over time.

```
arp -a  
192.168.56.1 -> 08:00:27:AA:AA:01  
192.168.56.1 -> 08:00:27:AA:AA:99 (changed)
```

Step 3. Review session logs - Search a sanitized access log for one token used by two source IPs.

```
session=LAB-SESSION-REDACTED  
192.168.56.20 21:10:01  
192.168.56.10 21:10:04
```

Step 4. Classify the event - Correlate the network and application evidence before raising an incident.

Detection output

The detector raised both a duplicate ARP warning and a same-session/different-source warning. The two signals together are stronger than either one alone.

Detection source	Evidence
Wireshark	Duplicate address detected for gateway IP 192.168.56.1
ARP table	Gateway MAC changed from ...AA:01 to ...AA:99
Application log	One redacted session token used from .20 and .10 within 3 seconds
Snort/Xarp	Training alert: possible ARP spoofing / MITM
Disposition	Confirmed simulated hijacking indicators; isolate and rotate sessions

Result

Session hijacking was detected successfully from both network and application perspectives. The recommended response is to isolate the suspected host, invalidate sessions, and investigate the ARP changes.

Safety and Submission Notes

- Do not publish raw tokens or full packet captures.
- Use time synchronization so cross-source correlation is reliable.
- Treat MAC changes as indicators, not proof, until corroborated.

Viva Questions and Answers

Why duplicate-IP warnings?

Two devices are claiming ownership of the same IP address in ARP replies.

Network vs application detection?

Network detection sees packet/ARP anomalies; application detection sees suspicious token or behavior changes.

What does arp -a show?

The local IP-to-MAC address cache.

What is Snort?

A network intrusion detection/prevention tool that matches traffic against rules.

How can session hijacking be prevented?

Use TLS, secure cookie flags, short lifetimes, reauthentication for sensitive actions, and network protections.